

NeuroHome: An Engineering Solution for Elderly Home Monitoring

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Outline

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- Other solutions
- Proposed Engineering Solution
- System Design and Architecture
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- Planning and Team Work
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Introduction

- Engineering solutions impact people's safety and daily life.
- Many elderly people live alone and face risks at home.
- NeuroHome proposes an engineering system to support elderly home monitoring.



Source: Traininghouse.pt

Problem context

Loneliness in elderly people is a problem:

- In Portugal, 2.5 million people are elderly.
- 40% of people with more than 80 years live alone.
- Therefore, this group is suitable to develop mental health problems.
- Support this elderly is a big challenge.



Source: [Traininghouse.pt](https://www.traininghouse.pt)

Problem Definition

NeuroHome want to give an answer for the following questions:

- How to monitor elderly people at home safely?
- How to detect abnormal situations?
- How to avoid intrusive or complex solutions?



Source: CareHop

Other solutions

There are other systems that try to act in the same problem:

Name	Description	Disadvantages
Dusun IOT	Montion, gas and temperature sensors for safety	Expensive and requires professional installation
ISmarch	Smart watch with vital sign detection and GPS	Needs to be charged and must be used by the elderly
Vayyar Home	Radar for fall detection and presence monitoring	Only focus on fall detection
Nursing homes	Place where people take care of the elderly	Expensive and limited offer

Engineering Requirements

The goals for our solution are:

- The system must be non-intrusive for elderly users.
- The system must be reliable and work continuously.
- Sensors must have low energy consumption.
- The system must be easy to use for elderly people and caregivers.
- Data must be secure and protected.

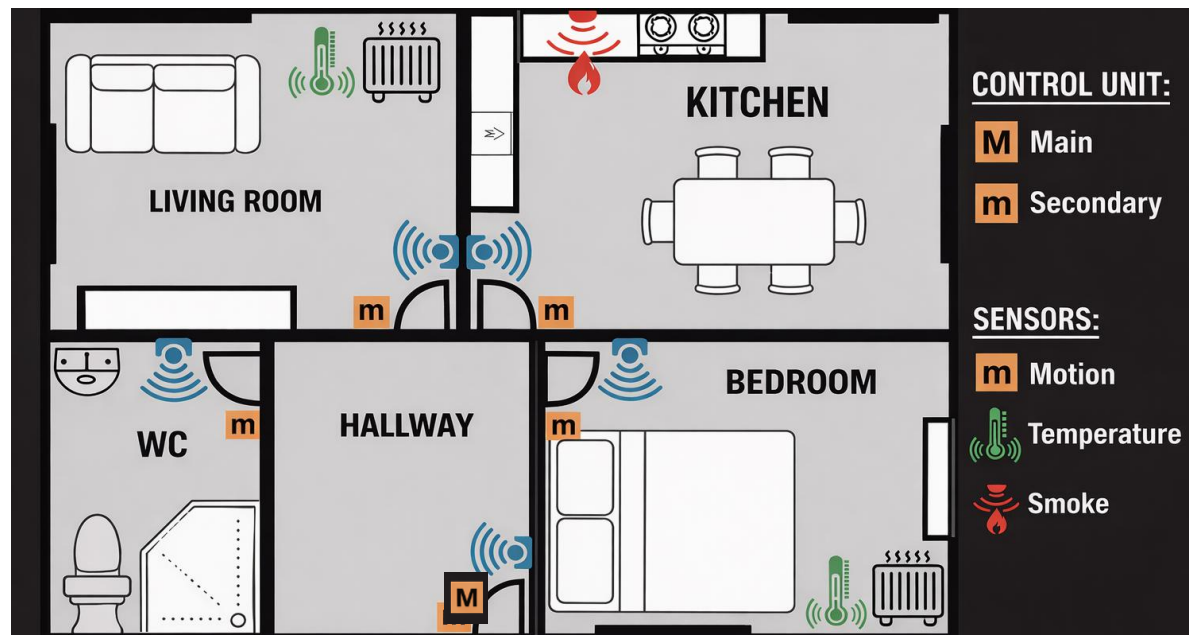


Source: FreePink

Proposed Solution – NeuroHome

NeuroHome is a home monitoring system:

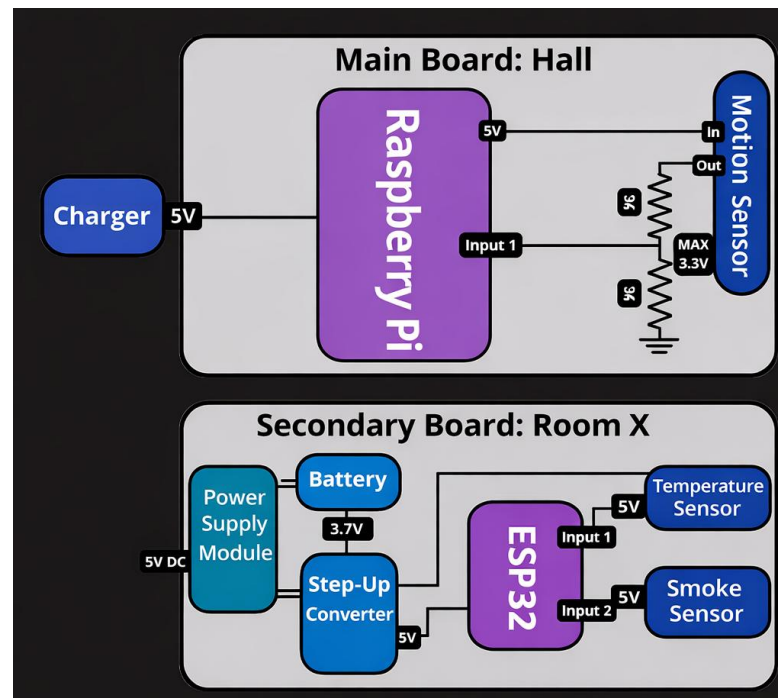
- Non-intrusive sensors are installed in the house.
- Sensors monitor movement, temperature and smoke.
- The system detects abnormal situations and sends alerts



System Architecture

The systems works in the following way:

- Sensors send data to the control unit.
- Data is processed locally and in the cloud.
- Alerts are generated and sent to caregivers.



Sensors and Data Collection

The data is collected by:

- Motion sensors that detect movement in different rooms.
- Temperature sensors that monitor environmental conditions.
- Smoke sensors that detect potential fire or gas risks.

The sensors collect data without using cameras or wearables.



Source: CareHop

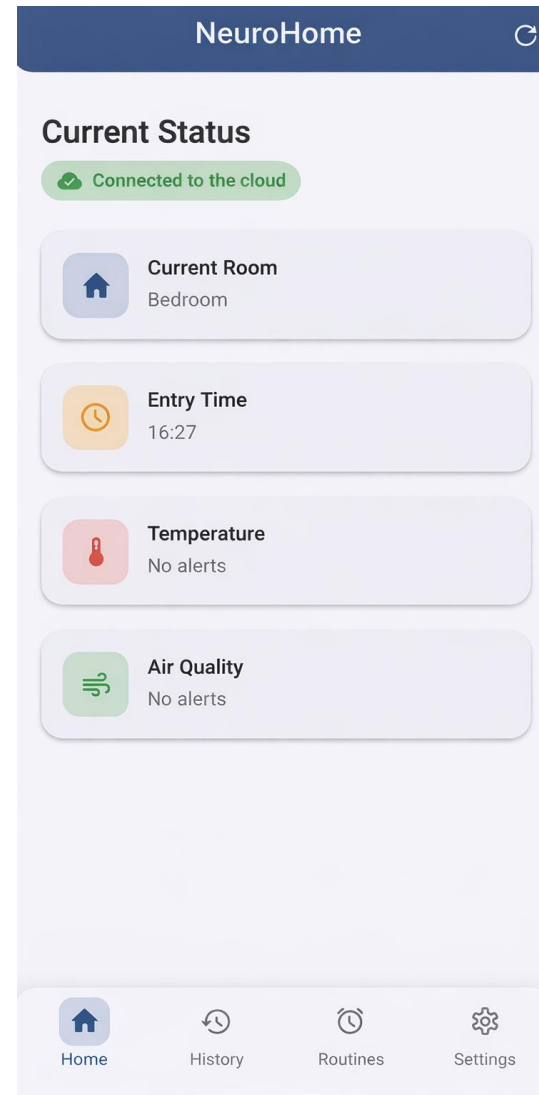
Data Processing and Anomaly Detection

- Sensor data is continuously collected.
- Normal daily routines are identified over time.
- The system detects deviations from normal behaviour.
- Abnormal patterns trigger alerts.



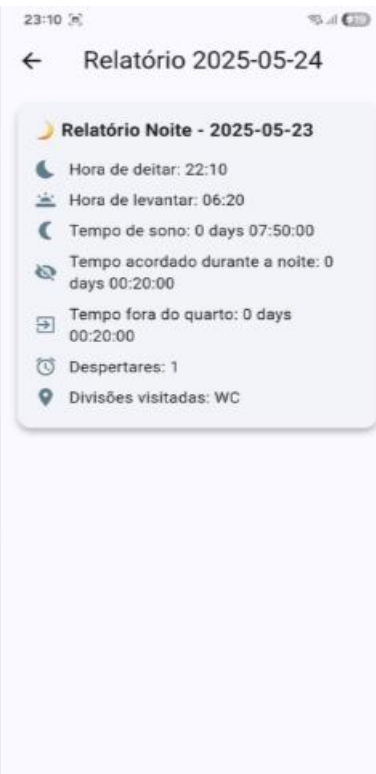
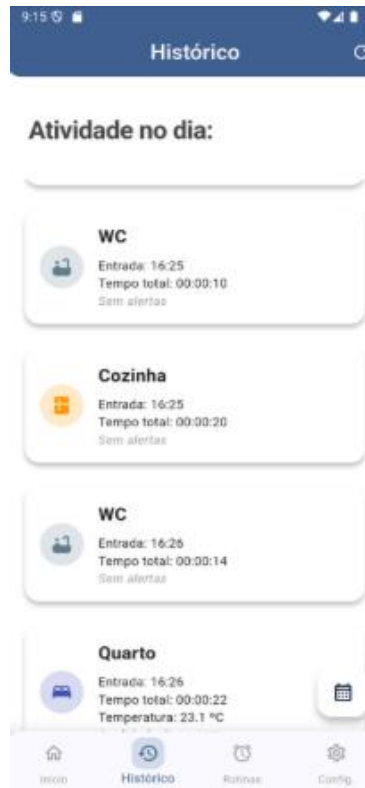
Alerts and User Interface

1. Abnormal situations trigger automatic alerts.
2. Alerts are sent to caregivers or family members.
3. Notifications allow quick response and intervention.



Alerts and User Interface

- App also keeps a history of the rooms where the elderly was.
- Twice a day it generates a report



Privacy and Ethical Considerations

- The system uses non-intrusive sensors only.
- No cameras or audio recording are used.
- Personal data is protected and anonymised.
- The solution respects users' privacy and dignity.



Source: LinkedIn

Advantages of NeuroHome

- Non-intrusive and easy to install.
- Continuous monitoring without user interaction.
- Early detection of risky situations.
- Supports independent living for elderly people.



Source: Griswold Home Care

Limitations and Challenges

- Sensor failures may affect system accuracy.
- False alerts can occur in some situations.
- Internet connectivity is required for full functionality.
- The system does not replace human care.



Source: Griswold Home Care

Future Improvements

- Integration of additional sensors (fall detection).
- Improved anomaly detection algorithms.
- Better personalization of user routines.
- Expansion to other types of users.



Source: The Auditor Outline

Impact and Contribution

- Improves safety for elderly people living alone.
- Supports caregivers and family members.
- Promotes independent living.
- Applies engineering to a real social problem.



Source: PapayaCare

Work Distribution

Person	Work
Beatriz Coelho	Develop routines detection algorithm, configuration of the Raspberry-pi, monitoring and testing
Francisco Castanheira	Develop the algorithm of the sensor, detection and configuration of ESP32, monitoring and testing
João Medeiros	Develop cloud system and mobile application, monitoring and testing

Conclusions

- NeuroHome addresses a real engineering and social problem.
- The solution is non-intrusive, reliable and user-friendly.
- Engineering decisions were guided by safety and ethics.
- The project demonstrates a complete engineering process.

References

- www.ine.pt/www.pordata.pt/www.ec.europa.eu/eurostat

Thank you

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